Move Two Bits, Create Four Networks

Explain it in your own words

Use this guide to understand each animation, choose your phrasing, and keep a natural YouTube delivery. Every clip has a matching frame, the idea behind the movement, and a specific delivery cue.

16 clips | 09:38.5 continuous video | 10 guide pages

A simple speaking rhythm

1. Set up: Tell viewers what the objects represent and ask one small question.

2. Follow the change: Point to what moves, grows, disappears or gets replaced.

3. Explain the cause: Say why that change happened. Use the mechanism notes as your checklist.

4. Let it land: Pause on the result, then connect it to the next question.

Use the timeline flexibly

The next page gives exact clip boundaries in the current continuous MP4. Individual clip files begin at 00:00. In the self-paced player, a phase holds until you advance, so your own recording can be longer or shorter. These timestamps describe the video; delivery cues describe how you might explain it.

Press Right / Left to move between phases, Up / Down to adjust speed, R to replay, and Shift+P for the separate script window. Rehearse the cause-and-effect in your own words before recording.

Open this video in the local player | Full narration reference | Sources

The complete timeline

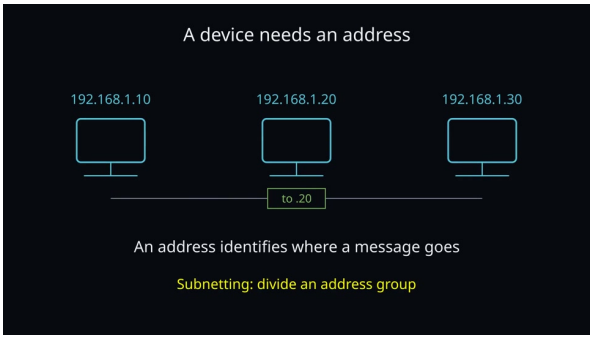
Find the clip, see the question being explained, and jump to its guide page. Times are MM:SS.t in the full MP4.

CLIP / TIME	WHAT YOU ARE EXPLAINING	PAGE
01 00:00.0 - 00:34.8	A device needs an address	3
02 00:34.8 - 01:10.4	Start with one tiny choice	3
03 01:10.4 - 01:45.6	Keep the beginning the same	4
04 01:45.6 - 02:22.2	Count the possible endings	4
05 02:22.2 - 02:58.7	Split the group in half	5
06 02:58.7 - 03:34.4	Split each half again	5
07 03:34.4 - 04:09.6	Count what each group holds	6
08 04:09.6 - 04:46.6	Find where each group starts	6
09 04:46.6 - 05:23.6	Make a stencil for the boundary	7
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13 07:11.9 - 07:48.9	Finish the address map	9
14 07:48.9 - 08:25.0	Choose enough room for devices	9
15 08:25.0 - 09:01.5	Check whether two devices match	10
16 09:01.5 - 09:38.5	One boundary explains the numbers	10

Use the clip number as your anchor when recording. Matching frames show the completed diagram; watch the movement too, because the change is often the explanation.

CLIP 01 / A device needs an address

00:00.0 - 00:34.8 in full video | 34.8s clip



What the viewer sees
An addressed message moves among office computers.

What you need to explain - and why
Devices need addresses so messages can be directed. Subnetting divides a collection of addresses into smaller groups.

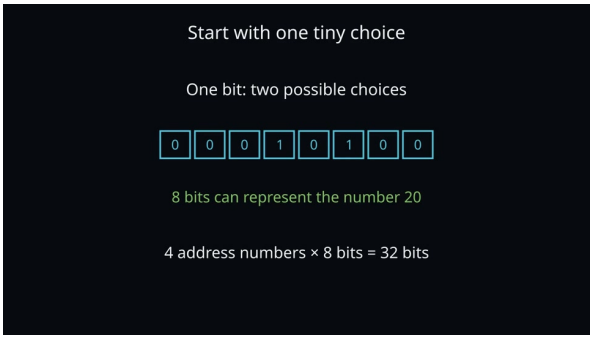
Tone and emphasis
Start with house-number intuition, then show the dotted address.

Next: Start with one tiny choice

Open clip 01 | My phrasing: _____

CLIP 02 / Start with one tiny choice

00:34.8 - 01:10.4 in full video | 35.7s clip



What the viewer sees
One 0/1 choice grows into eight bits and a 32-bit address.

What you need to explain - and why
A bit has two choices. Eight bits represent one dotted IPv4 number; four such groups contain 32 bits.

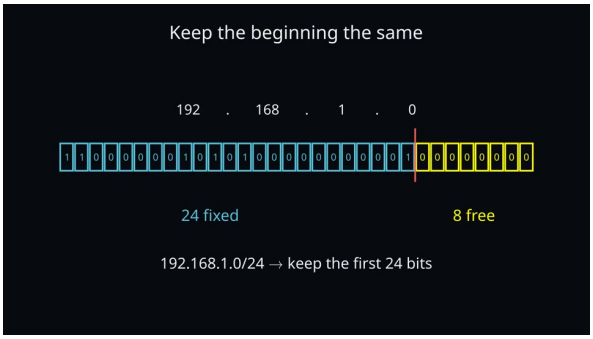
Tone and emphasis
Introduce the choices before showing the long binary row.

Next: Keep the beginning the same

Open clip 02 | My phrasing: _____

CLIP 03 / Keep the beginning the same

01:10.4 - 01:45.6 in full video | 35.2s clip



What the viewer sees
Several addresses share a beginning, then a boundary marks 24 bits.

What you need to explain - and why
A /24 prefix fixes the first 24 bits and leaves eight free. Prefix means the beginning all addresses in this block share.

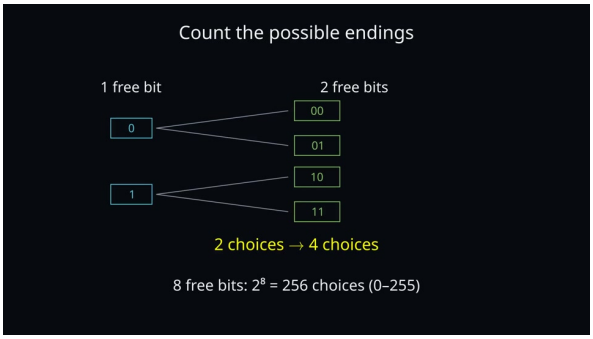
Tone and emphasis
Translate the slash aloud as how many beginning bits stay fixed.

Next: Count the possible endings

Open clip 03 | My phrasing: _____

CLIP 04 / Count the possible endings

01:45.6 - 02:22.2 in full video | 36.5s clip



What the viewer sees
One free bit branches into two-bit combinations.

What you need to explain - and why
Each extra free bit doubles the number of possibilities. Eight free bits give 256 endings, numbered zero through 255.

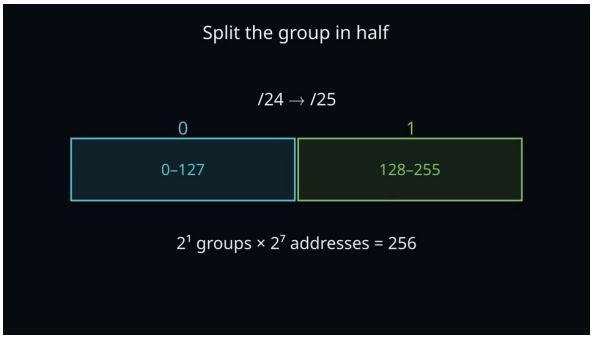
Tone and emphasis
Explain why zero counts and why the final value is 255 rather than 256.

Next: Split the group in half

Open clip 04 | My phrasing: _____

CLIP 05 / Split the group in half

02:22.2 - 02:58.7 in full video | 36.5s clip



What the viewer sees
One block splits into two equal halves.

What you need to explain - and why
Use one formerly free bit to identify a half. Seven bits remain free inside each /25 group, giving 128 addresses.

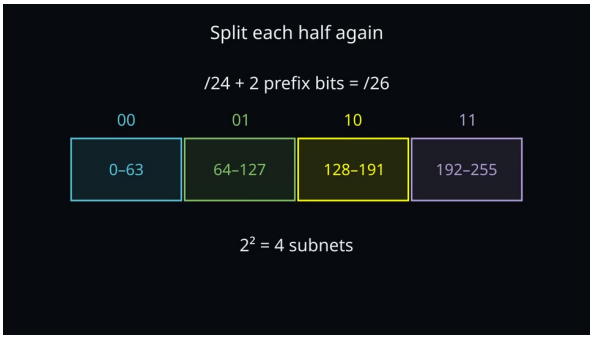
Tone and emphasis
Keep the total collection constant while showing the new boundary.

Next: Split each half again

Open clip 05 | My phrasing: _____

CLIP 06 / Split each half again

02:58.7 - 03:34.4 in full video | 35.7s clip



What the viewer sees
Each half splits again into four labeled groups.

What you need to explain - and why
Two group bits have four combinations: 00, 01, 10, 11. The shared beginning is now 26 bits, giving four /26 subnets.

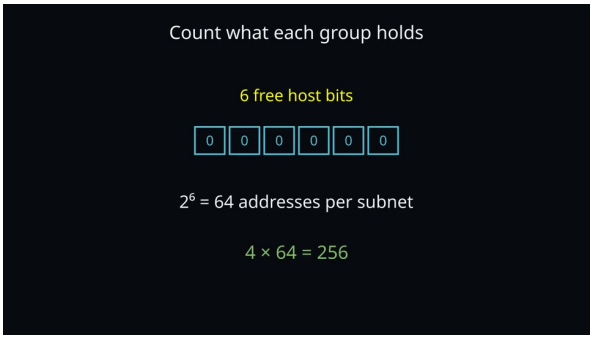
Tone and emphasis
Use two yes-or-no choices before naming four subnet prefixes.

Next: Count what each group holds

Open clip 06 | My phrasing: _____

CLIP 07 / Count what each group holds

03:34.4 - 04:09.6 in full video | 35.2s clip



What the viewer sees
Six remaining free bits are counted.

What you need to explain - and why
Six host bits give 64 addresses inside each group. Four groups times 64 exactly accounts for the original 256.

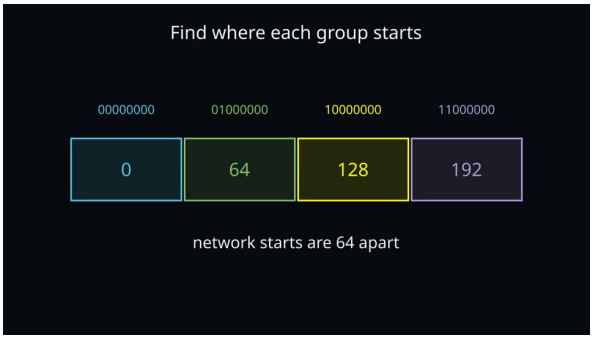
Tone and emphasis
Distinguish total addresses from addresses available to individual devices.

Next: Find where each group starts

Open clip 07 | My phrasing: _____

CLIP 08 / Find where each group starts

04:09.6 - 04:46.6 in full video | 37.0s clip



What the viewer sees
Four subnet starts appear at equal gaps.

What you need to explain - and why
The starts are 0, 64, 128 and 192 in the last dotted number. Each block occupies 64 places, so the next starts 64 later.

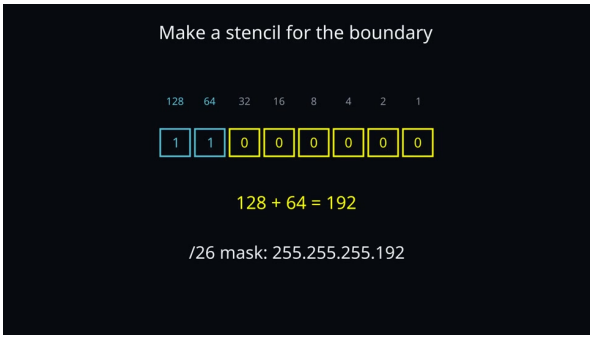
Tone and emphasis
Build starts by counting block sizes rather than memorizing numbers.

Next: Make a stencil for the boundary

Open clip 08 | My phrasing: _____

CLIP 09 / Make a stencil for the boundary

04:46.6 - 05:23.6 in full video | 37.0s clip



What the viewer sees
A mask marks shared positions with ones.

What you need to explain - and why
For /26, the final eight mask bits are 11000000. Their place values add to 128 + 64 = 192, giving mask 255.255.255.192.

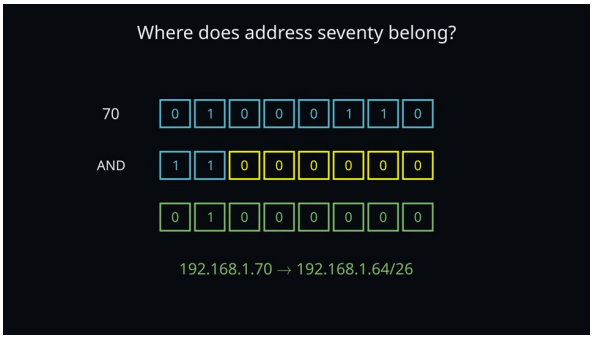
Tone and emphasis
Describe a stencil that keeps the shared beginning.

Next: Where does address seventy belong?

Open clip 09 | My phrasing: _____

CLIP 10 / Where does address seventy belong?

05:23.6 - 05:58.3 in full video | 34.8s clip



What the viewer sees
The bits for seventy are placed above the mask.

What you need to explain - and why
AND keeps address bits where the mask is one and writes zero elsewhere. 70 masked by 192 gives 64, identifying the containing subnet.

Tone and emphasis
Walk left to right through the retained and erased bits.

Next: Two addresses have special jobs

Open clip 10 | My phrasing: _____

CLIP 11 / Two addresses have special jobs

05:58.3 - 06:35.3 in full video | 37.0s clip



What the viewer sees
The first and last addresses receive special roles.

What you need to explain - and why
In this ordinary local-network example, the first names the subnet and the last is broadcast. 64 total minus two special addresses leaves 62 for devices.

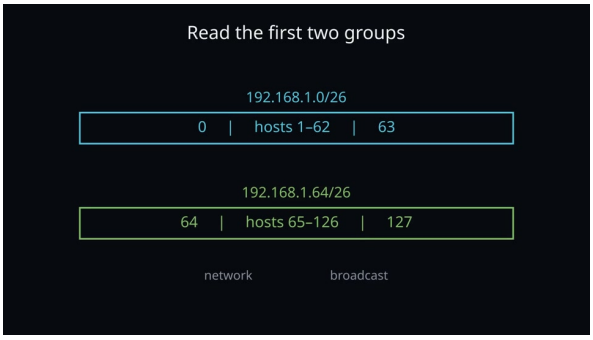
Tone and emphasis
Keep the ordinary-LAN qualification; special arrangements use other rules.

Next: Read the first two groups

Open clip 11 | My phrasing: _____

CLIP 12 / Read the first two groups

06:35.3 - 07:11.9 in full video | 36.5s clip



What the viewer sees
The first two blocks display their endpoints and device ranges.

What you need to explain - and why
0-63 has devices 1-62; 64-127 has devices 65-126. The broadcast address belongs to this block's end, not always .255.

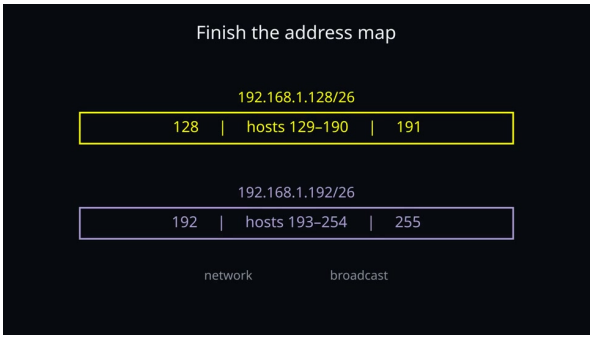
Tone and emphasis
Point to endpoints before reading the middle ranges.

Next: Finish the address map

Open clip 12 | My phrasing: _____

CLIP 13 / Finish the address map

07:11.9 - 07:48.9 in full video | 37.0s clip



What the viewer sees
The remaining blocks complete the original collection.

What you need to explain - and why
128-191 and 192-255 complete the four blocks with no gaps or overlaps. Each reserves its own first and last address.

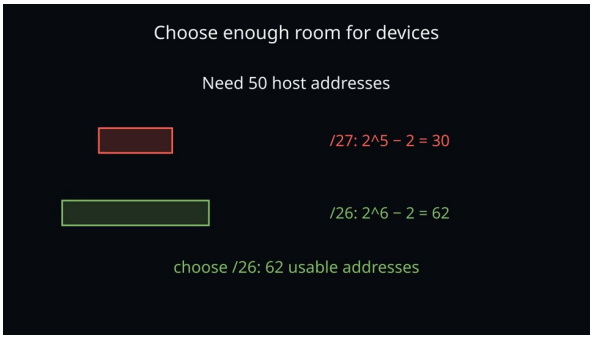
Tone and emphasis
Let viewers verify that the bars cover the whole original block.

Next: Choose enough room for devices

Open clip 13 | My phrasing: _____

CLIP 14 / Choose enough room for devices

07:48.9 - 08:25.0 in full video | 36.1s clip



What the viewer sees
Candidate sizes are compared with a need for fifty devices.

What you need to explain - and why
Five free bits leave 30 ordinary device addresses, which is too small. Six leave 62, so /26 provides enough room.

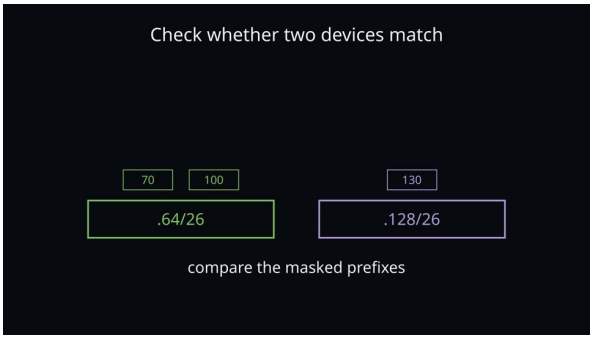
Tone and emphasis
Choose required room first, then derive the slash number.

Next: Check whether two devices match

Open clip 14 | My phrasing: _____

CLIP 15 / Check whether two devices match

08:25.0 - 09:01.5 in full video | 36.5s clip



What the viewer sees
Addresses seventy, one hundred and one hundred thirty enter groups.

What you need to explain - and why
Under /26, 70 and 100 share the prefix starting at 64; 130 belongs to the prefix starting at 128.

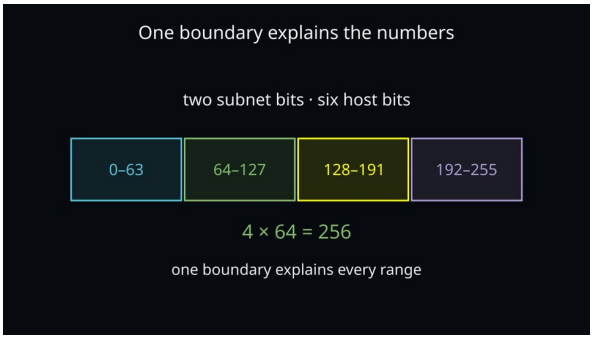
Tone and emphasis
Show why matching the first three dotted numbers alone is insufficient here.

Next: One boundary explains the numbers

Open clip 15 | My phrasing: _____

CLIP 16 / One boundary explains the numbers

09:01.5 - 09:38.5 in full video | 37.0s clip



What the viewer sees
Four groups and the movable boundary return.

What you need to explain - and why
Subnet starts, masks and address counts all follow from deciding which bits identify a group and which vary inside it.

Tone and emphasis
End on one collection divided by a boundary, not a list of isolated rules.

Close: return to the opening question and its answer.

Open clip 16 | My phrasing: _____